

Learner Name

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1. Did you use any AI tool(s) to assist with this assignment?

☐ No. Why not?

☒ Yes. List the AI tool(s) you used and briefly describe the purpose for which you used the tool(s). (e.g., brainstorming, research, drafting content)

I used Claude in generating a synthetic proxy dataset matching my Module 2 Data Dictionary schema, with realistic noise, since the real Kaggle dataset could not be pulled into the working environment. I also used it in writing and running the ingestion, cleaning, transformation, validation, bias detection, and anonymization scripts; building and running the Prefect DAG and pytest suite; writing the Dockerfile, CI workflow, and governance docs.

2. To demonstrate effective AI use, please provide your initial prompt(s) and a brief description of the steps you took to refine and iterate on the AI's output for this

Initial prompt: I uploaded the Module 3 brief and asked Claude to build a working data pipeline for my FirstBank project, continuing from Modules 1-2, with all required artifacts (DAG, validation suite, unit tests, Dockerfile, bias detection, anonymization, governance, audit logging) actually built and run, not just described.

I refined it by doing the following:

1. Confirmed the injected data messiness was realistic 2. Ran the pytest suite and caught a real case-sensitivity bug in the cleaning logic, fixed and re-verified it 3. Checked the Great Expectations and bias detection results against the actual data 4. Confirmed the DAG genuinely ran end-to-end via the log 5. Noted the Dockerfile could not be build-tested here (no Docker daemon) and flagged this for me to verify myself.

3. List at least one source you consulted to verify the AI-generated content in your work.

- Great Expectations' official documentation (docs.greatexpectations.io) to confirm the expectation types used are current API
- Fairlearn's official documentation (fairlearn.org) to confirm the representation-parity approach used in the bias detection suite is a standard, accepted method.